Figures and Tables Style Guide

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One source gives the paper's figure, the website's figure, and the slide's figure. Talks have their own leeway (the talks guide). Agents follow `.setup/ai/style/figures-tables.md`, which follows this guide and is changed together with it.

1 Pipeline

Kind	Source	Paper	Web
Result plot	`Fig<Name>()` in `R`	PDF	PNG
Diagram	`.setup/latex/tikz/<name>.tex`	PDF	PNG
Table	results in `.setup/data-*`, columns by name	`booktabs`	`knitr::kable()`
Animation	`Fig<Name>()`, one frame per call	small multiples	GIF

2 Plots

- Palette: Okabe-Ito (`grDevices::palette.colors()`), with a point shape and a line type per group, so that figures read in greyscale.
- Format: vector PDF for papers and slides, PNG at 300 dpi for pkgdown, Quarto, and the README; both from the same source.
- A `Fig<Name>()` is exported, documented, and tested. It saves the files itself, at a size preset, in `.setup/latex/figures`.

- A `highlight` or `subset` argument is optional, for a figure likely to be shown in stages: each stage is one call, shown with `\only<n>{}` in Beamer.

`size`	Width × height (in)	Use
`"column"`	3.3 × 2.5	one column of `jou`; body-text font
`"page"`	6.5 × 4	text width; body-text font
`"slide"`	5.5 × 3.1	16:9 frame; larger sans-serif font

The height can be overridden. LaTeX sets `width = \linewidth` or `\textwidth`, never `scale`.

3 Diagrams

Diagrams are standalone TikZ files in `.setup/latex/tikz`. `make latex` compiles each to PDF, then `pdftoppm -r 300` makes the PNG; each is rebuilt only when its source is newer.

4 Animations

- GIFs are allowed everywhere: pkgdown, Quarto, README, Reveal.js, Beamer, and papers.
- `GifWrite(fig, frames, fps, size)` calls a `Fig*()` once per frame and writes `.setup/latex/figures/gif/<nam` the PNG frames, and a multi-page PDF, using `magick`.
- Papers show small multiples: a few chosen frames as one faceted PDF figure. A paper may also embed the animation with the `animate` package (it plays only in Acrobat and Okular), always next to the small multiples, never in place of them.
- Under 10 MB per GIF; 2 to 10 frames per second; loops.
- No flashing faster than 3 per second. Every GIF has alt text and a caption that says what changes. Meaning is never shown by colour alone.

5 Tables

- Read results from `.setup/data-*` and select columns by name, never by number.
- `.Tabular(x = x, digits = 3)` writes the rows; `\rinline{}` puts them in a hand-written `booktabs` and `threeparttable` frame.
- The web version is `knitr::kable()` from the same data.

6 Floats

- APA 7: the caption above, as a title in title case; `\textit{Note.}` below.
- Every float and subfigure has a label, `fig:<project>-<name>` or `tab:<project>-<name>`.
- Every figure chunk has a label, since an unnamed chunk's file name shifts when a chunk is added.

7 Helpers

The template writes copies, with no dependency on rProject: `.setup/r-miscellaneous/texTools` (`.Tabular()`, `.Round()`, `.Array()`, the biblatex helpers, `GifWrite()`, `GifFrames()`) with the `ggplot2` theme and palette; a package with `Fig*()` moves them into `R` and imports `ggplot2`.

8 Tools

`make lint` reports (never fails on) `scale` and missing files in `\includegraphics`, floats without a caption or label, figure chunks without a label, GIFs over 10 MB or without a source, and papers that embed an animation without small multiples.

References

R Core Team. (2026). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. Vienna, Austria. https://www.R-project.org/